University
Mohammed VI
Polytechnic

College of
Computing

Lab 6: SQL – DDL and DML

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Member 1	BENYASSINE Rayane
Member 2	Name: AIT LAHCEN Adam
Member 3	Name: BITAR Driss
Repository Link	https://github.com/Adam-Ait-Lahcen/Base-de-donn-es

1 Introduction

In this lab, we implemented the relational schema of the Research Lab Management System (RLMS) using SQL. The work consists of defining the database structure through DDL, populating and modifying its content through DML, and translating relational queries into SQL. It also allows us to examine the limits of the integrity constraints expressed during the implementation.

2 Team Roles

Parts 1 & 2 (DDL & Schema Evolution, DML): Carried out collaboratively by Adam AIT LAHCEN and Driss BITAR. For Part 1, after discussing the migration strategy, we initially sought to retain the existing schema and apply modifications using ALTER TABLE queries; however, because the existing schema contained numerous unnamed constraints, dropping and altering them was error-prone, prompting us to adjust the source script directly. In Part 2, Driss BITAR authored all DML statements, while the database population and sequence validation were performed jointly by Adam AIT LAHCEN and Driss BITAR.

Part 3 (SQL Queries): The query formulation and implementation were carried out entirely by Driss BITAR. Adam AIT LAHCEN was responsible for running tests, verifying query results against expected tuples, keeping the execution log, and documenting comprehensive review notes regarding query layout and potential optimizations.

Part 4 & Report Synthesis: Rayane BENYASSINE was primarily responsible for investigating the integrity constraints that could not be enforced strictly through standard declarative SQL constructs, as well as synthesizing, editing, and compiling the final lab report.

3 Main Implementation Difficulties

The key technical difficulties arose during Parts 1 and 2 of the implementation: Part 1 (DDL & Unnamed Constraint Handling): When evolving the relational schema from previous labs, our initial plan was to write procedural SQL alteration scripts (ALTER TABLE) to modify tables without altering base files. However, we encountered multiple unnamed integrity and foreign key constraints in the existing database schema. Because the DBMS auto-generated internal system identifiers for these constraints, selectively dropping and updating them via scripts became unwieldy and fragile. We therefore scrapped that approach and modified the DDL code directly. Furthermore, we had to carefully calibrate primary and composite keys, NULL attributes, and referential actions such as ON DELETE CASCADE and NOT ON DELETE SET NULL . Part 2 (DML & Dependency Sequencing): Populating the tables required rigorous attention to referential integrity. Due to inter-table foreign key dependencies, inserting and updating records had to follow an exact dependency order so that referenced primary key records existed prior to foreign key assignment. Part 3 (SQL Query Translation): Translating the relational algebra formalisms into SQL queries was straightforward, relying on multi-table joins, nested subqueries, grouping with aggregations, and set-theoretic operations developed in prior labs.

4 An answer for the open question of Part 4

While translating the RLMS relational schema into SQL, we were able to express several integrity constraints using primary keys, foreign keys, and NOT NULL declarations. However, these constructs are not sufficient to enforce every requirement of the application. Some constraints go beyond the validity of individual attributes or references between tables, since they depend on the existence of certain relationships, the number of records satisfying a given condition, or the consistency of information distributed across several tables. We can illustrate these limitations through the following requirements:

PE6 – Each student must be supervised by at least one academic staff member: The Advises table represents the supervision relationship between students and academic staff, but its primary and foreign keys only ensure that the recorded relationships are valid, not that every student has at least one advisor. The database would therefore need to verify the existence of at least one corresponding record in Advises for each student, particularly when a new student is added or an existing supervision relationship is deleted.

PR3 – Each research project must have exactly one Principal Investigator (PI): The Participates table records the role of each person in a research project, and its composite primary key (pid, projectCode) ensures that a person cannot have multiple participation records in the same project. However, this does not guarantee that each project has exactly one PI, so the database would need to verify that exactly one participant is assigned this role and that this participant belongs to Academic, particularly when a project is created or its participation records are inserted, updated, or deleted.

CE7 – Only people holding the required certifications are allowed to use or reserve equipment units: The Requires and Holds tables represent the certifications required by an equipment model and those held by a person, but their foreign keys do not guarantee that the person making a reservation satisfies these requirements. The database would need to compare the certifications required by each reserved equipment model with those held by the person, taking into account their validity at the planned time of use, when-ever equipment is added to a reservation or the relevant reservation and certification information is modified.